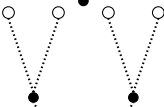


Renormalized equation

$$(\partial_t - \Delta) u = u_x^2 + \xi$$

with $\xi \in C^{-3/2-\kappa}$, parabolic scaling $\mathfrak{s} = (2, 1)$, spatial dimension $d = 1$.

Divergent trees

τ	$ \tau $	$S(\tau)$	k_τ	$F(\tau^*)$
	$-3/2 - \kappa$	1	k_0	1
	$-1 - 2\kappa$	2	k_2	2
	$-1/2 - 3\kappa$	2	k_3	4
	$-1/2 - \kappa$	1	–	0
	$-1/2 - \kappa$	1	k_1	$2u_x$
	-4κ	8	k_6	8
	-4κ	2	k_7	8
	-2κ	1	–	0
	-2κ	2	–	0

τ	$ \tau $	$S(\tau)$	k_τ	$F(\tau^*)$
	-2κ	1	k_4	$4u_x$
	-2κ	2	k_5	$4u_x$

Legend: $\circ$ = noise, $\bullet$ = integration node, solid edge = $\mathcal{I}$, dotted edge = $\partial\mathcal{I}$ (derivative), X^n above a node = polynomial (Taylor) decoration, e.g. X_x ; recall $(\Pi X^n)(y) = y^n$.

Renormalized family

$$\begin{aligned}
(\partial_t - \Delta) u &= u_x^2 + \xi \\
&+ k_0 \\
&+ k_2 \\
&+ 2k_3 \\
&+ 2k_1 u_x \\
&+ k_6 \\
&+ 4k_7 \\
&+ 4k_4 u_x \\
&+ 2k_5 u_x
\end{aligned}$$

Canonical (BPHZ) renormalization

Each free constant at its canonical value $k_\tau = h(S'_-\tau)$ for a centered Gaussian noise. Mean-zero parity makes trees with an odd number of noise vertices vanish (3 of 8 below); the survivors are polynomials in the elementary expectations $h_i \equiv h_\varepsilon(\sigma) = \mathbb{E}[\Pi^\varepsilon \sigma](0)$ of the ε -regularized noise ξ_ε .

These constants are ε -dependent and diverge as $\varepsilon \rightarrow 0$; the integrals below are left unevaluated.

$$k_0 = 0 \quad (\text{vanishes: odd noise parity})$$

$$k_2 = -h_0$$

$$k_3 = 0 \quad (\text{vanishes: odd noise parity})$$

$$k_1 = 0 \quad (\text{vanishes: odd noise parity})$$

$$k_6 = 2h_0^2 h_2 h_8 - h_0^2 h_7 - 2h_0 h_1 h_8 - 2h_0 h_2 h_6 + 2h_0 h_5 + 2h_1 h_6 - h_4$$

$$k_7 = h_0 h_{10} - h_0 h_{11} h_2 - h_0 h_3 h_8 + h_1 h_{11} + 2h_{11} h_3 + h_3 h_6 - h_9$$

$$k_4 = -h_3$$

$$k_5 = h_0 h_2 - h_1$$

giving the canonically renormalized equation(s)

$$\begin{aligned} (\partial_t - \Delta) u &= u_x^2 + \xi \\ &+ -h_0 \\ &+ 2h_0^2 h_2 h_8 - h_0^2 h_7 - 2h_0 h_1 h_8 - 2h_0 h_2 h_6 + 2h_0 h_5 + 2h_1 h_6 - h_4 \\ &+ 4h_0 h_{10} - 4h_0 h_{11} h_2 - 4h_0 h_3 h_8 + 4h_1 h_{11} + 8h_{11} h_3 + 4h_3 h_6 - 4h_9 \\ &+ -4h_3 u_x \\ &+ 2h_0 h_2 u_x - 2h_1 u_x \end{aligned}$$

where, for the ε -regularized noise ξ_ε (covariance C_ε) and the singular kernel K of L^{-1} , the elementary expectations are given below. An entry marked = 0 vanishes, and $(= h_j)$ marks a duplicate (same value up to the o -decoration).

$$h_0 = h_\varepsilon \left(\begin{array}{c} \circ \quad \circ \\ \diagdown \quad \diagup \\ \bullet \end{array} \right)$$

$$h_0 = \int_{(\mathbb{R}^2)^2} C_\varepsilon(z_1 - z_2) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_2) \, dz_1 \, dz_2$$

$$h_1 = h_\varepsilon \left(\begin{array}{c} \circ \quad \circ \\ \diagdown \quad \diagup \\ \bullet \\ \vdots \\ \bullet \end{array} \right)$$

$$h_1 = \int_{(\mathbb{R}^2)^3} C_\varepsilon(z_2 - z_3) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(z_1 - z_2) \partial^{(0,1)} K(z_1 - z_3) \, dz_1 \, dz_2 \, dz_3$$

$$h_2 = h_\varepsilon \left(\begin{array}{c} o = -1 - 2\kappa \\ \square \\ \vdots \\ \bullet \end{array} \right)$$

$$h_2 = \int_{(\mathbb{R}^2)^1} \partial^{(0,1)} K(-z_1) \, dz_1$$

= 0 — pure-kernel total derivative

$$h_3 = h_\varepsilon \left(\begin{array}{c} \circ \\ \vdots \\ \bullet \quad \circ \\ \vdots \\ \bullet \end{array} \right)$$

$$h_3 = \int_{(\mathbb{R}^2)^3} C_\varepsilon(z_2 - z_3) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_3) \partial^{(0,1)} K(z_1 - z_2) \, dz_1 \, dz_2 \, dz_3$$

$$h_4 = h_\varepsilon \left(\begin{array}{c} \circ \quad \circ \quad \circ \quad \circ \\ \vdots \quad \vdots \quad \vdots \quad \vdots \\ \bullet \quad \bullet \\ \vdots \\ \bullet \end{array} \right)$$

$$\begin{aligned} h_4 = & \int_{(\mathbb{R}^2)^6} C_\varepsilon(z_2 - z_3) C_\varepsilon(z_5 - z_6) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_4) \\ & \partial^{(0,1)} K(z_1 - z_2) \partial^{(0,1)} K(z_1 - z_3) \partial^{(0,1)} K(z_4 - z_5) \partial^{(0,1)} K(z_4 - z_6) \, dz_1 \, dz_2 \, dz_3 \, dz_4 \, dz_5 \, dz_6 \\ & + \int_{(\mathbb{R}^2)^6} C_\varepsilon(z_2 - z_5) C_\varepsilon(z_3 - z_6) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_4) \\ & \partial^{(0,1)} K(z_1 - z_2) \partial^{(0,1)} K(z_1 - z_3) \partial^{(0,1)} K(z_4 - z_5) \partial^{(0,1)} K(z_4 - z_6) \, dz_1 \, dz_2 \, dz_3 \, dz_4 \, dz_5 \, dz_6 \\ & + \int_{(\mathbb{R}^2)^6} C_\varepsilon(z_2 - z_6) C_\varepsilon(z_3 - z_5) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_4) \\ & \partial^{(0,1)} K(z_1 - z_2) \partial^{(0,1)} K(z_1 - z_3) \partial^{(0,1)} K(z_4 - z_5) \partial^{(0,1)} K(z_4 - z_6) \, dz_1 \, dz_2 \, dz_3 \, dz_4 \, dz_5 \, dz_6 \end{aligned}$$

$$h_5 = h_\varepsilon \left(\begin{array}{c} \circ \quad \circ \\ \vdots \quad \vdots \\ \bullet \quad \bullet \\ \vdots \\ \bullet \end{array} \right) \quad \begin{array}{c} o = -1 - 2\kappa \\ \text{red square} \end{array}$$

$$h_5 = \int_{(\mathbb{R}^2)^4} C_\varepsilon(z_2 - z_3) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_4) \partial^{(0,1)} K(z_1 - z_2) \partial^{(0,1)} K(z_1 - z_3) \, dz_1 \, dz_2 \, dz_3 \, dz_4$$

$$h_6 = h_\varepsilon \left(\begin{array}{c} \circ \quad \circ \\ \vdots \quad \vdots \\ \bullet \\ \vdots \\ \text{red square} \end{array} \right) \quad \begin{array}{c} o = -2\kappa \\ o = +2\kappa \end{array}$$

$$h_6 = \int_{(\mathbb{R}^2)^3} C_\varepsilon(z_2 - z_3) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(z_1 - z_2) \partial^{(0,1)} K(z_1 - z_3) \, dz_1 \, dz_2 \, dz_3$$

$$= h_1 \text{ (same value)}$$

$$h_7 = h_\varepsilon \left(\begin{array}{c} o = -1 \quad o = 1 - 2\kappa \\ \text{red square} \quad \text{red square} \\ \diagdown \quad \diagup \\ \bullet \end{array} \right)$$

$$h_7 = \int_{(\mathbb{R}^2)^2} \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_2) \, dz_1 \, dz_2$$

$$= 0 \text{ — pure-kernel total derivative}$$

$$h_8 = h_\varepsilon \left(\begin{array}{c} o = -1 - 2\kappa \\ \text{red square} \\ \vdots \\ o = 2\kappa \\ \text{red square} \end{array} \right) \quad (o = -2\kappa)$$

$$h_8 = \int_{(\mathbb{R}^2)^1} \partial^{(0,1)} K(-z_1) \, dz_1$$

$$= 0 \text{ — pure-kernel total derivative}$$

$$h_9 = h_\varepsilon \left(\begin{array}{c} \circ \quad \circ \\ \diagdown \quad \diagup \\ \bullet \\ \diagdown \quad \diagup \\ \bullet \quad \circ \\ \diagdown \quad \diagup \\ \bullet \quad \circ \\ \diagdown \quad \diagup \\ \bullet \end{array} \right)$$

$$\begin{aligned} h_9 = & \int_{(\mathbb{R}^2)^6} C_\varepsilon(z_3 - z_4) C_\varepsilon(z_5 - z_6) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_6) \\ & \partial^{(0,1)} K(z_1 - z_2) \partial^{(0,1)} K(z_1 - z_5) \partial^{(0,1)} K(z_2 - z_3) \partial^{(0,1)} K(z_2 - z_4) \, dz_1 \, dz_2 \, dz_3 \, dz_4 \, dz_5 \, dz_6 \\ & + \int_{(\mathbb{R}^2)^6} C_\varepsilon(z_3 - z_5) C_\varepsilon(z_4 - z_6) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_6) \\ & \partial^{(0,1)} K(z_1 - z_2) \partial^{(0,1)} K(z_1 - z_5) \partial^{(0,1)} K(z_2 - z_3) \partial^{(0,1)} K(z_2 - z_4) \, dz_1 \, dz_2 \, dz_3 \, dz_4 \, dz_5 \, dz_6 \\ & + \int_{(\mathbb{R}^2)^6} C_\varepsilon(z_3 - z_6) C_\varepsilon(z_4 - z_5) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_6) \\ & \partial^{(0,1)} K(z_1 - z_2) \partial^{(0,1)} K(z_1 - z_5) \partial^{(0,1)} K(z_2 - z_3) \partial^{(0,1)} K(z_2 - z_4) \, dz_1 \, dz_2 \, dz_3 \, dz_4 \, dz_5 \, dz_6 \end{aligned}$$

$$h_{10} = h_\varepsilon \left(\begin{array}{c} o = -1 - 2\kappa \\ \text{red square} \quad \circ \\ \diagdown \quad \diagup \\ \bullet \quad \circ \\ \diagdown \quad \diagup \\ \bullet \end{array} \right)$$

$$\begin{aligned}
 h_{10} = \int_{(\mathbb{R}^2)^4} C_\varepsilon(z_3 - z_4) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_4) \partial^{(0,1)} K(z_1 - z_2) \\
 \partial^{(0,1)} K(z_1 - z_3) \, \mathrm{d}z_1 \, \mathrm{d}z_2 \, \mathrm{d}z_3 \, \mathrm{d}z_4
 \end{aligned}$$

$$h_{11} = h_\varepsilon \Big(\overset{o=}{\underset{2\kappa}{\textcolor{red}{\square}}} \begin{array}{c} \circ \\ \vdots \\ \bullet \end{array} \circ \Big)$$

$$\begin{aligned}
 h_{11} &= \int_{(\mathbb{R}^2)^3} C_\varepsilon(z_2 - z_3) \partial^{(0,1)} K(-z_1) \partial^{(0,1)} K(-z_3) \partial^{(0,1)} K(z_1 - z_2) \, \mathrm{d}z_1 \, \mathrm{d}z_2 \, \mathrm{d}z_3 \\
 &= h_3 \text{ (same value)}
 \end{aligned}$$